

Usability

LG Architecture Training Program

Paulo Merson

Agenda

A white rectangular card is placed on a wooden desk. On the desk, there are several writing tools: a blue pen, a silver pen, a white pen, a pair of white compasses, a small box of pens, and a blue eraser. The card has the following text:

➔ Usability intro
Usability tactics

Usability

- Measurable quality that involves:
 - How easy it is to learn how to operate the system
 - Efficiency when using the system (time to complete a task)
 - Handling of mistakes made by the user (how many mistakes; irritability upon mistakes)
 - User confidence that the system is OK (frequent and visible feedback)
 - Satisfaction in using the system
 - Accessibility



*Ergonomics and UX
(user experience) are
terms related to
usability*

User experience (UX) goes beyond usability

Usability

- Make a task easy and intuitive
- Minimize steps and remove barriers
- What users do, how they do it
- Focus on the system's UI

UX

- Make a task significant and valuable
- Create an emotional connection
- What users feel
- Scope is beyond the UI (includes for example human interaction)

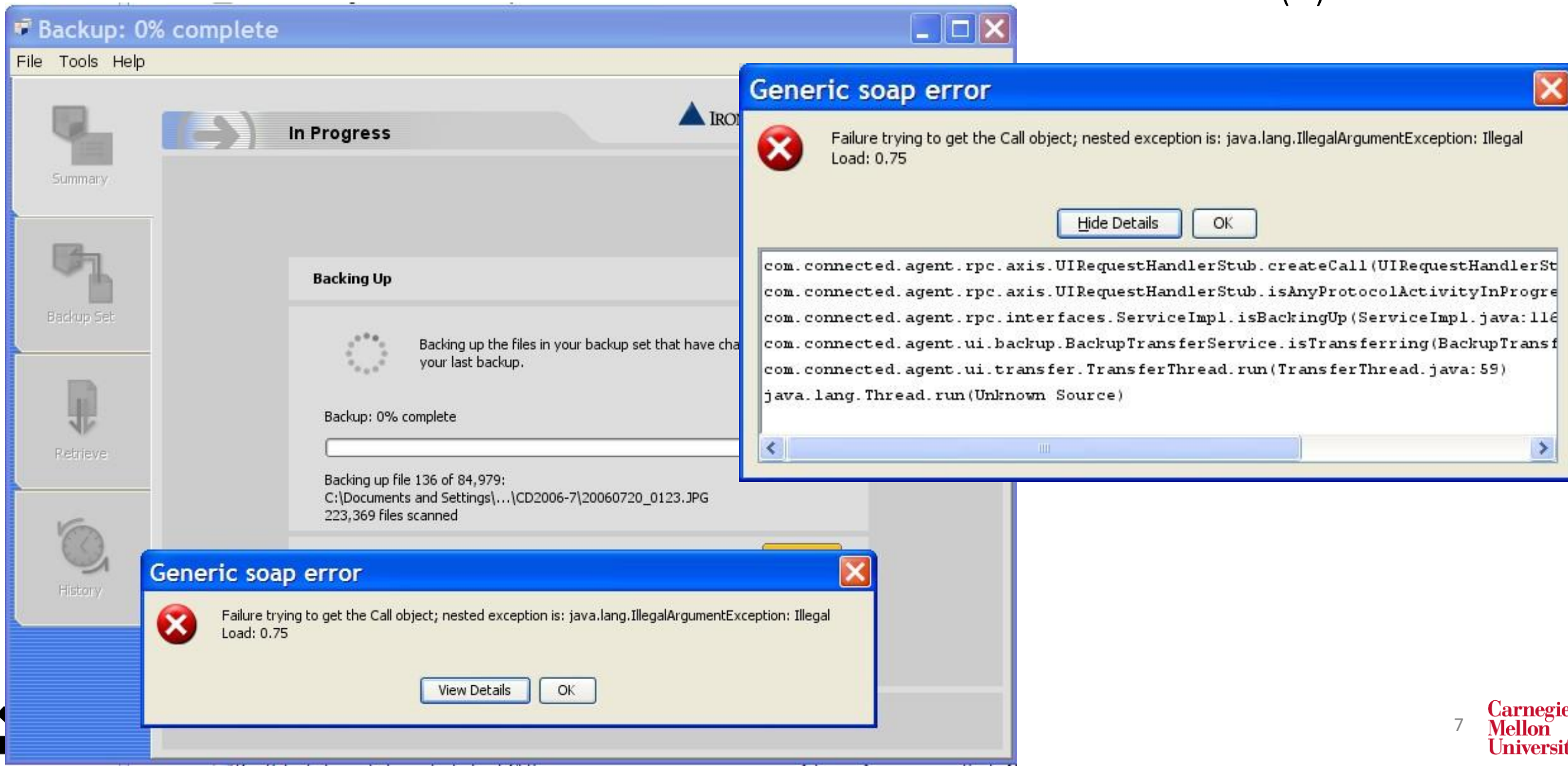
Guideline #1: error messages

- There are two audiences for error messages:
 - **Developers** are interested in debug info (stack trace, module name and line, snapshot metrics, etc.)
 - **End users** need to know what to do
 - Try again later
 - Call support assistance
 - Restart the device/computer
- And hence two channels
 - For developers: log records, message headers/bodies, core dumps, etc.
 - For end users: the user interface, email, push notifications

Error messages through the wrong channel (1)



Error messages through the wrong channel (2)



Error messages through the wrong channel (3)

The screenshot shows a Mozilla Firefox browser window with the address bar displaying `http://www.usair.com/awa/content/dividendmiles/purchasemiles/buymiles.aspx`. The page title is "US Airways | Buy Miles". The browser's search bar shows "us airways" delta - Google News. The page content includes a navigation bar with links like "Travel Planning", "Dividend Miles", "Specials", "Travel Tools & Services", and "About US". A sidebar on the left lists "Dividend Miles" links such as "Join Now", "My Dividend Miles", "Book Award Travel", "Member Benefits", "All About Preferred", "Earn More Miles", and "US Airways Credit Cards". The main content area displays an error message: "Error: The following error(s) occurred when processing your request." followed by three Oracle error codes: "ORA-06510: PL/SQL: unhandled user-defined exception ORA-06512: at 'SYS.UTL_HTTP', line 174", "ORA-06512: at 'AMWEST.HPCONNECTOR', line 100 ORA-06512: at 'AMWEST.HPVALIDATE', line 229", and "ORA-06512: at line 1". A "Go Back" button is visible below the error message. At the bottom of the page, there is a "POWERED BY points-c-m" logo and a copyright notice: "Copyright 2006 US Airways Inc. All Rights Reserved.".

US Airways | Buy Miles - Mozilla Firefox

File Edit View Go Bookmarks Tools Help

http://www.usair.com/awa/content/dividendmiles/purchasemiles/buymiles.aspx

Go sita atlanta

"us airways" delta - Google News

US Airways | Buy Miles

Corporate profile

Welcome PAULO F MERSON : 92,597 | Log Out

U.S AIRWAYS Fly with us.

U.S AIRWAYS DIVIDEND MILES PREFERRED

Getting to Preferred status is easy when you fly First Class and Go for Gold! Learn More

Travel Planning | Dividend Miles | Specials | Travel Tools & Services | About US

Home > Dividend Miles > BUYMILES

Error: The following error(s) occurred when processing your request.

ORA-06510: PL/SQL: unhandled user-defined exception ORA-06512: at "SYS.UTL_HTTP", line 174
ORA-06512: at "AMWEST.HPCONNECTOR", line 100 ORA-06512: at "AMWEST.HPVALIDATE", line 229
ORA-06512: at line 1

Go Back

Dividend Miles

- Join Now
- My Dividend Miles
- Book Award Travel
- Member Benefits
- All About Preferred
- Earn More Miles
- US Airways Credit Cards

Need Help?

POWERED BY points-c-m Get More Rewards, faster.

Copyright 2006 US Airways Inc. All Rights Reserved.

U.S AIRWAYS DIVIDEND MILES SHOPPING MALL

Error messages through the wrong channel (4)



Hello, Paulo

ESP

ENG

POR

Discover ConnectMiles

Earn Miles

Use Miles

PreferProgram

Service Center

License

Liferay Portal Enterprise Edition 6.2.10 EE GA1

Newton / Build 6210 / November 1, 2013

Server Info

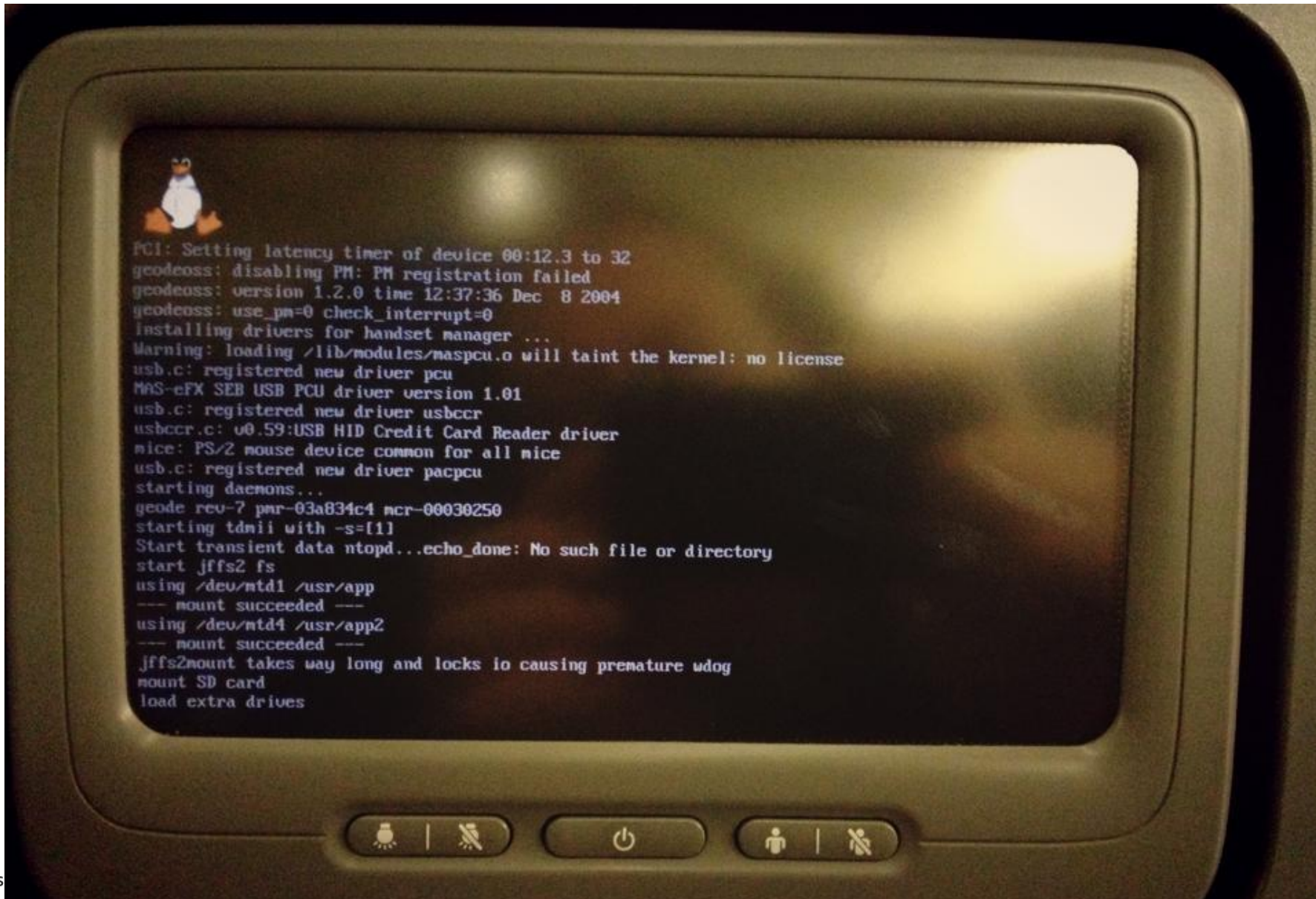
Licenses Registered

	Host Name	IP	MAC Addresses	Product	Status	Owner	Description	Type	Start Date	Expiration Date	Additional Information
	Addresses	Addresses	Addresses								
☐	cc6d02b9ca9f:8080	172.17.0.3	a6:b0:48:57:66:02	Portal Production (Additional JVM)	Expired	Liferay Trial	Connect Miles P Temp	production	9/6/2016	10/6/2016	
	Host Name	IP	MAC Addresses	Product	Status	Owner	Description	Type	Start Date	Expiration Date	Additional Information
	Addresses	Addresses	Addresses								
☐	337bc23f31b9:8080	172.17.0.3	a6:b0:48:57:66:03	Portal Production	Active	Compania Panamena de Aviación S.A.	Compania Panamena de Aviación S.A. Connect Miles P Def	production	8/29/2016	8/5/2116	

Register Your Server or Application

Order Id

Why passengers should see this?



Guideline #2: make the UI more efficient

- Smart and tolerant data entry
- Auto-complete
- Memory of data previously entered (see Maintain User Model tactic)
- Avoid unnecessary clicks and page reloads

UIs that could be more efficient ⁽¹⁾

The screenshot shows a Mozilla Firefox browser window titled "Polícia Federal - DPF - Mozilla Firefox". The address bar shows "http://www.dpf.gov.br/". The page header includes the "Ministério da Justiça" and "Departamento de Polícia Federal" logos and navigation links. A sidebar on the left lists various services like "Concursos", "Contas Públicas", etc. The main content area features a "Consultar Situação da Solicitação" form. This form has three input fields: "Número de Protocolo:" with the value "1.2007.0000093722", "CPF:" with the value "19807694173", and a CAPTCHA field with the text "orccbo". Below these fields are buttons for "Confirmar", "Fechar", and "Ajuda". An "Atualizar" link is also present. An error message dialog box is overlaid on the bottom of the form, stating: "The page at http://www.dpf.gov.br says: Apenas um dos parâmetros da pesquisa necessita ser preenchido".

Could search by protocol number or CPF

Error message complaining that both were entered

UIs that could be more efficient (2)



Welcome PAULO F MERSON
Hertz #1 Club Gold
Five Star Member

#1 Awards Point Balance.....3908
2009 Rental YTD.....3

[Logout](#)

My Profile >

My Account >

Reservations >

Special Offers >

Locations >

Vehicle Guide >

PERSONAL DATA

Hertz #1 Club Gold
Five Star

The Hertz Corporation, a Delaware (U.S.) corporation

- Driver's License - Driver's license expiration date day field must be two characters long. [NEX023]

PAULO F MERSON

CREATE YOUR OWN USER ID >

[Learn More](#)

Password:

••••

Re-enter Password:

••••

Phone: (At least one is required. Do not include dashes or spaces.)

Home Phone:

4127322793

Business Phone:

4127322793

Preferred Fax Number:

4122685758

Driver's License Number:

(do not include spaces or dashes)

7487

Country - State/Province:

United States - Pennsylvania

Driver's License Expiration Date:

Month Day Year

Feb

6

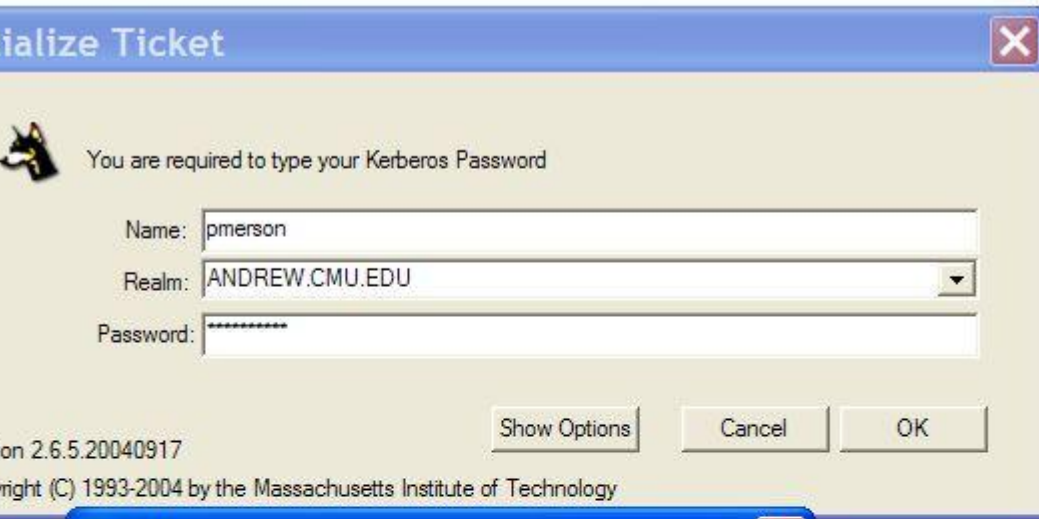
2013

Complaining that the day is not valid

Had to type 06 instead of 6



UIs that could be more efficient (3)



Initialize Ticket

You are required to type your Kerberos Password

Name: pmerson

Realm: ANDREW.CMU.EDU

Password: *****

Show Options Cancel OK

on 2.6.5.20040917
right (C) 1993-2004 by the Massachusetts Institute of Technology



Leash

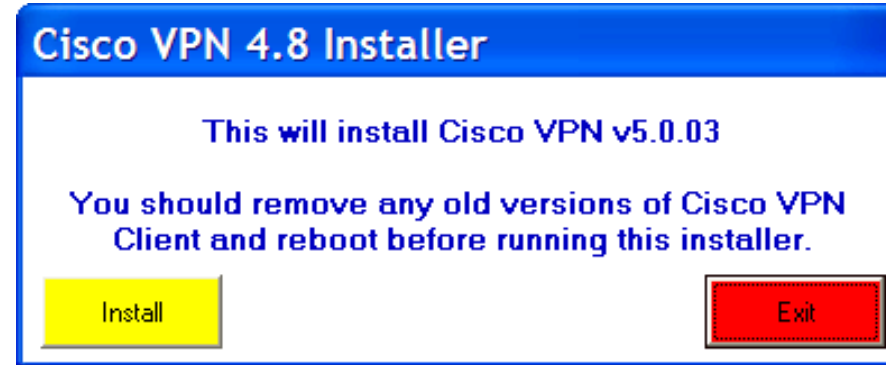
Ticket initialization failed.

Kerberos 5: Decrypt integrity check failed (error 31)
Kerberos 4: Password incorrect
(Kerberos error 62)

Password incorrect
(Kerberos error 62)

OK

*Why not just say
"incorrect
credentials"?*



Cisco VPN 4.8 Installer

This will install Cisco VPN v5.0.03

You should remove any old versions of Cisco VPN Client and reboot before running this installer.

Install Exit

*Why doesn't the
installer prompt me to
uninstall the previous
version or does so
automatically?*



Login de usuários fora da rede TCU - M

File Edit View Favorites Tools Help

Back Forward Stop Home Search

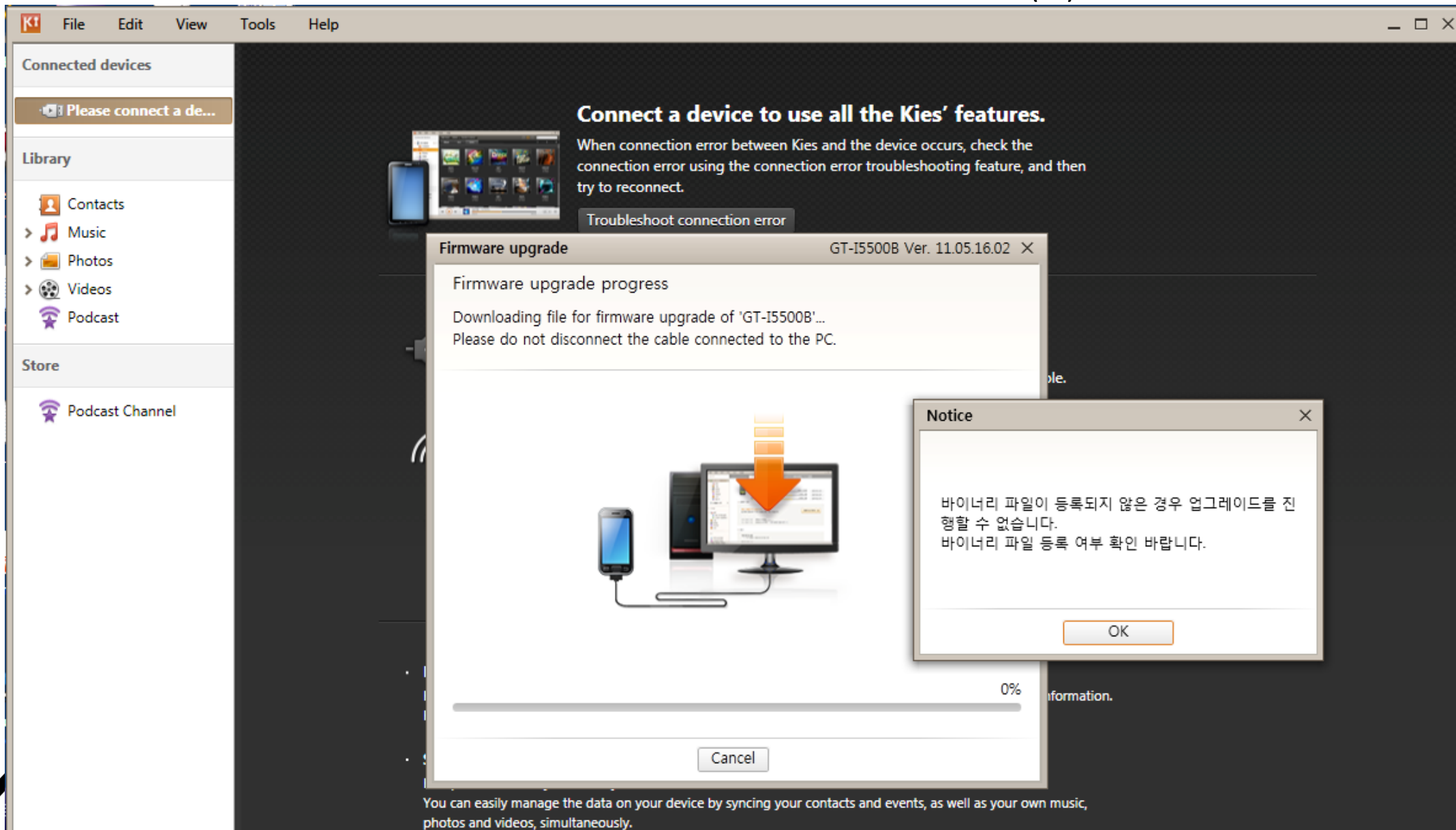
Address http://www2.tcu.gov.br/portal/page?_pageid=33,2

Login

Usuário: SIGA: Senha: Conectar

*Why not set focus to
user ID input?
onload... focus()*

UIs that could be more efficient (4)



Lost in translation



Agenda

Usability intro

➔ Usability tactics

Usability tactics

- Cancel
- Undo
- Pause/resume
- Aggregate
- Maintain user model

Usability is a broad field with many tactics, guidelines, analysis techniques, and more

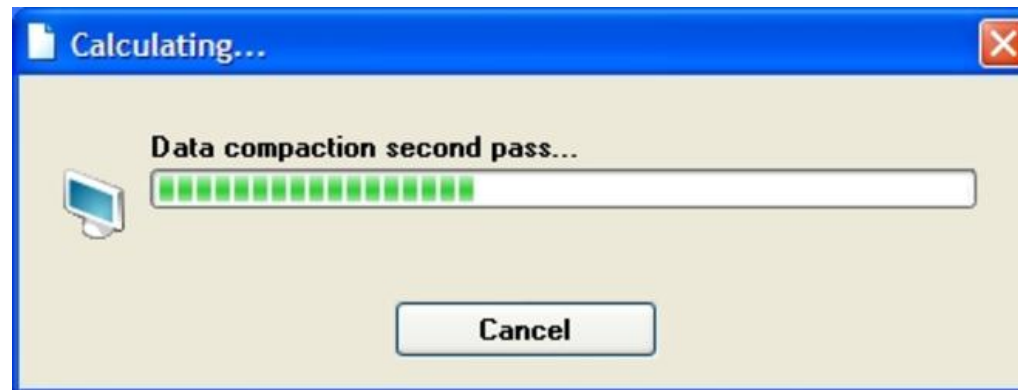


Jakob
Nielsen



Cancel

- Allows the user to cancel lengthy operations
- The architecture involves:
 - UI thread that waits for the cancel command, and another thread executing the lengthy operation
 - Lengthy operation checking if cancel was issued at each step
 - When cancel is activated:
 - Free resources
 - Close and rollback transactions
 - Communicate the cancelling to the user and other components



*We're talking about
threads that run on the
same machine/device*

Undo

- The user can undo the last operation (or up to N previous operations)
- The architecture involves:
 - Operation stack
 - A mechanism to restore or reconstruct state after undo, such as:
 - Snapshots
 - Saga pattern (for distributed systems)
 - Correction events (for event-driven architectures)
 - Command design pattern [Gamma94]
 - Memento design pattern [Gamma94]

*Some operations
are irreversible*

Pause/resume

- Allows the user to pause and resume long-running operations
- The architecture involves:
 - UI thread that waits for the pause and resume commands, and another thread that performs the lengthy operation
 - Free resources if possible when paused
 - Pause timeout in some cases
 - Operation state persistence in some cases (aka dehydration)



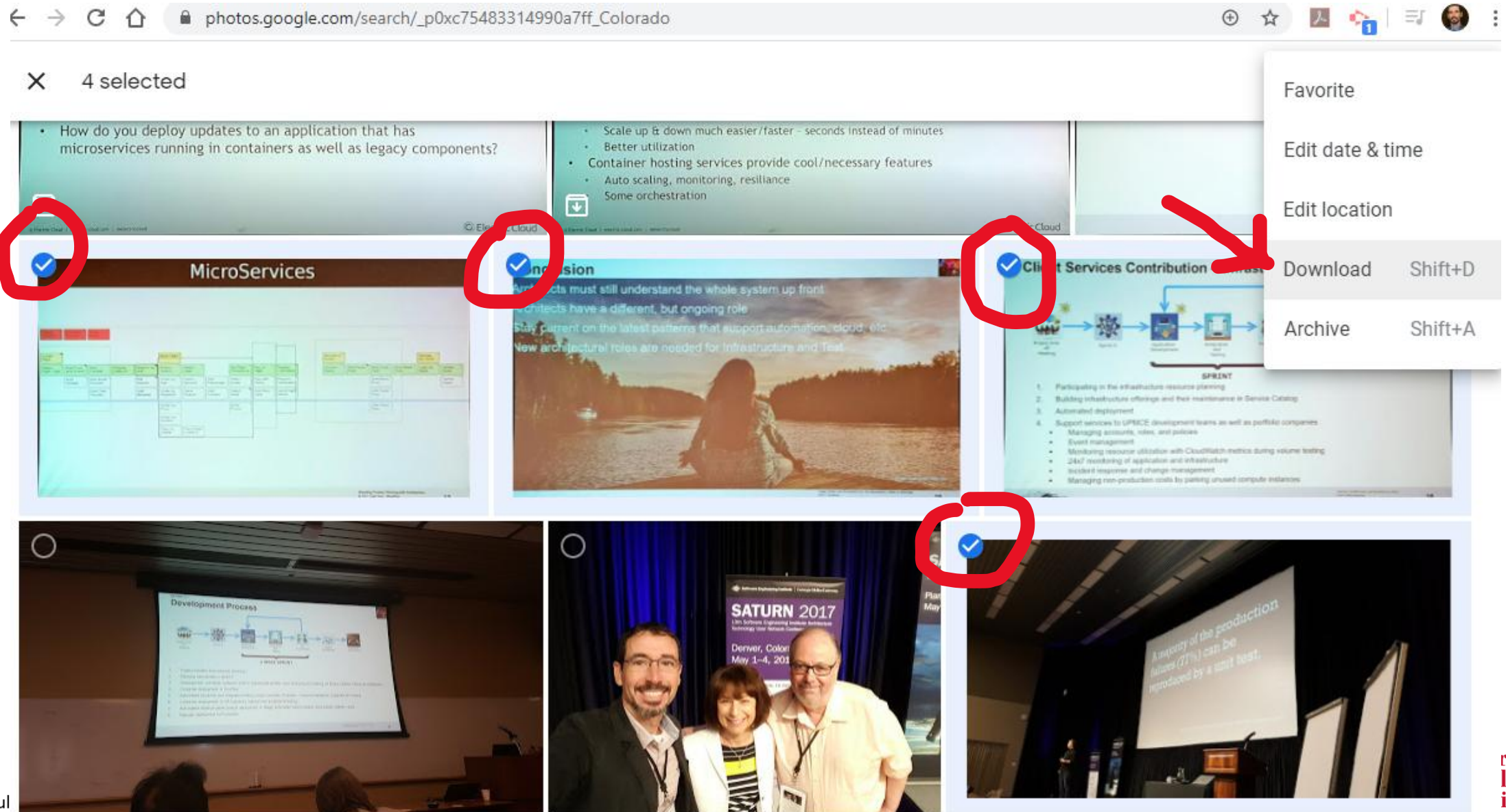
Aggregate (commands or data)

- The system executes together several **commands** usually performed together
- The architecture involves:
 - Composing commands in sequence
 - Defining the sequence of execution of composed commands
- Several **data elements** are combined as the input to an operation
- The architecture involves:
 - Multi-select in the UI
 - A component to manage collections of data elements

Aggregate commands example



Aggregate data example



Maintain user model

- The system maintains data about a specific user (or class of users)
- Examples of mechanisms:
 - UI customization
 - Memorize user activity to create a model of preferences and anticipate needs
- The architecture involves:
 - Data collection with minimal overhead for operations performed
 - UI option to configure user preferences with persistence mechanism
 - Machine learning model in some cases
 - Security to protect user data
 - Compliance to regulations (transparency and consent are usually required)



Usability and security: a common tradeoff



Software must be prepared for use and abuse

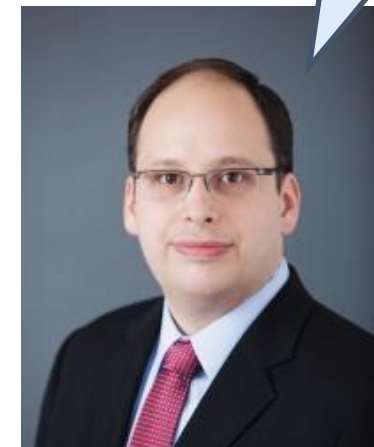


Leon Bambrick
@secretGeek

There are 2 hard problems in computer science: cache invalidation, naming things, and off-by-1 errors.

12:20 PM · Jan 1, 2010 · Twitter Web Client

Hugh Thompson





Questions?

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